Fast Fourier Transform (FFT)

IDFT

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Inverse DFT

- **Signal x[n] can be fully recovered from its N-point DFT X[k] provided N ≥ L (length of signal x[n]).**

- **Inverse DFT is used to get x[n] from X[k]*:***

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}, n = 0,1, \dots N-1$$

- **Take complex conjugate on both sides:**

$$x^*[n] = \frac{1}{N}\sum_{k=0}^{N-1} X^*[k]e^{-j\frac{2\pi}{N}kn}, n = 0,1, \dots N-1$$

- **Compare with DFT:**

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \dots, N-1$$

Inverse DFT Using FFT

Need to get signal x[n] from DFT X[k], given:

$$x^*[n] = \frac{1}{N}\sum_{k=0}^{N-1} X^*[k]e^{-j\frac{2\pi}{N}kn}, n = 0,1, \dots N-1 \text{ (IDFT)}$$

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}, k = 0, 1, \dots, N-1 \text{ (DFT)}$$

Compute IDFT using the following steps:

- Take complex conjugate of X[k] to get $X^*[k]$.
- Compute N-point FFT of $X^*[k]$.
- Divide result by N.
- Take complex conjugate of result to get x[n].

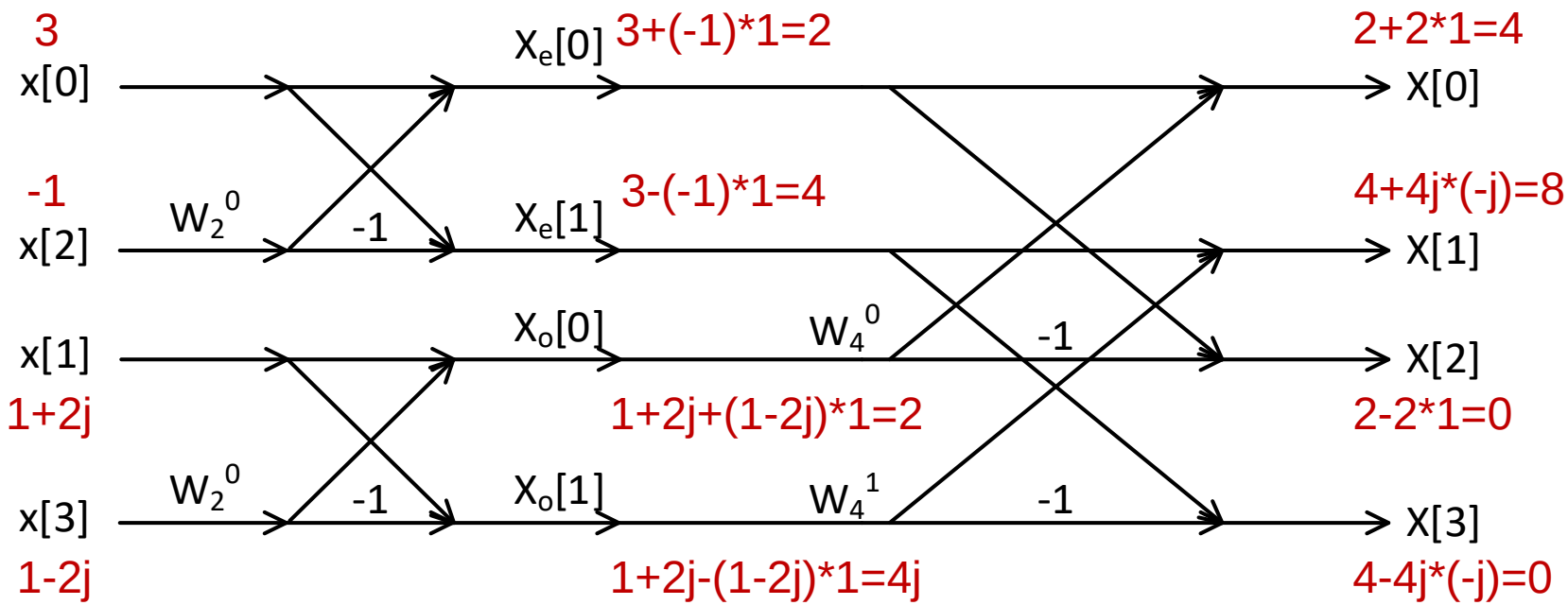
Example

- **Sketch the FFT flow graph to get 4-point signal x[n] from 4-point DFT X[k] = [3, 1-2j, -1, 1+2j].**

Answer

$$W_N^k = e^{-j\frac{2\pi}{N}k}$$
$$W_2^0 = 1$$
$$W_4^0 = 1$$
$$W_4^1 = -j$$

- X[k] = [3, 1-2j, -1, 1+2j]
- Take complex conjugate X*[k] = [3, 1+2j, -1, 1-2j] and do FFT:



- Divide result [4 8 0 0] by N = 4, we get [1 2 0 0].
- Take complex conjugate to get x[n] = [1 2 0 0].

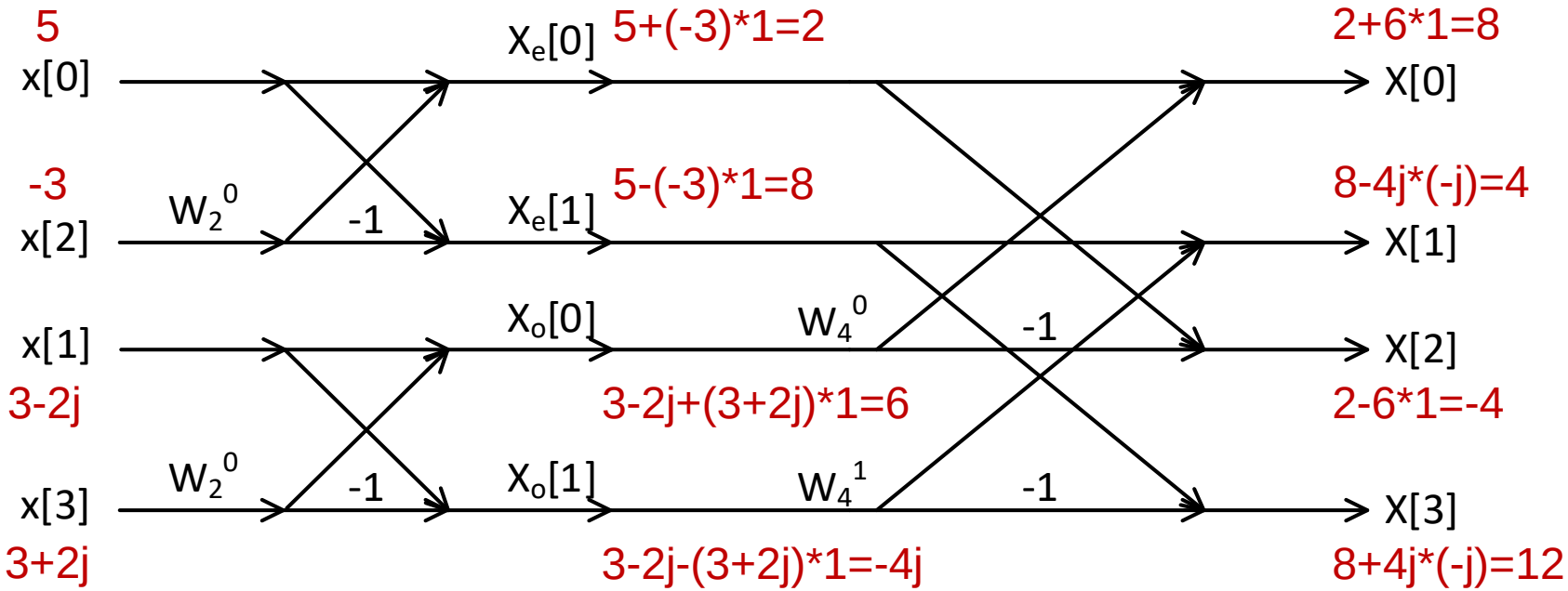
Exercise

- **Sketch the FFT flow graph to get 4-point signal x[n] from 4-point DFT X[k] = [5, 3+2j, -3, 3-2j].**

Answer

$$W_N^k = e^{-j\frac{2\pi}{N}k}$$
$$W_2^0 = 1$$
$$W_4^0 = 1$$
$$W_4^1 = -j$$

- X[k] = [5, 3+2j, -3, 3-2j]
- Take complex conjugate X*[k] = [5, 3-2j, -3, 3+2j] and do FFT:



- Divide result [8 4 -4 12] by N = 4, we get [2 1 -1 3].
- Take complex conjugate to get x[n] = [2 1 -1 3].

Question

Take complex conjugate of DFT X[k] = [1, 2-j, -3, 2+j] and arrange it in bit reverse order. What vector do you get?

a) [1, 2-j, -3, 2+j]

b) [1, 2+j, -3, 2-j]

c) [1, -3, 2+j, 2-j]

d) [1, -3, 2-j, 2+j]

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